

Pranesh Nur

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SKILLS

Generative AI	Stable Diffusion, ControlNet, LoRA Fine-Tuning, IP-Adapter, HuggingFace Diffusers
ML / Deep Learning	PyTorch, Multimodal Learning, NumPy, Pandas, OpenCV
MLOps & Deployment	MLflow, FastAPI, Docker, Nginx, ONNX, GitHub Actions, Pytest, Git LFS, Google Cloud Run, Firebase
Systems & Languages	Python, C++, C, Bash, SQL, CMake, OOP, Data Structures
Dev Environment	Neovim, fzf, Git, Jupyter Notebook

PROJECTS

PneumoDetect: End-to-End Pneumonia Detection System *[Live Demo]*

- **Tech Stack:** Python, PyTorch, EfficientNetV2-S, ONNX, FastAPI, Docker, Nginx, MLflow, GitHub Actions, Pytest, Git LFS, Google Cloud Run, Firebase Hosting
- Deployed full-stack medical AI system to **Google Cloud Run** (backend) and **Firebase Hosting** (frontend) with automated **CI/CD** via **GitHub Actions** - tests on every push, deployment on merge to main; **zero errors** across 10,000 production requests.
- Implemented two-stage inference pipeline: CNN-based **guard model** rejects non-X-ray inputs **7.2× faster** than full pipeline (P99 320ms vs 2,303ms); **EigenCAM** explainability via ONNX intermediate layer extraction and PCA projection - no external CAM library.
- Trained on [Kermany et al. \(Cell, 2018\)](#) - 5,856 clinically validated pediatric X-rays; exported **EfficientNetV2-S** to **ONNX** achieving **96.15% test accuracy** and **1.54× inference speedup** (97ms → 63ms), tracked end-to-end with **MLflow**.
- Optimised production Docker image **93.8% from 4.1GB to 256MB** by removing PyTorch and torchvision from inference container - reducing cold start from 34s to 8s; benchmarked warm inference across 10,000 requests: **P50 1.2s, P99 2.3s**.

A Benchmark of Visual Attention Mechanisms in CNNs

- **Tech Stack:** Python, PyTorch, MLflow, argparse, Jupyter Notebook
- Engineered a reproducible benchmarking pipeline comparing **SE** and **CBAM** attention mechanisms against a ResNet baseline across four datasets - achieving **+10.61% accuracy** improvement with CBAM on **Oxford102Flowers**.
- Tracked all experiments with **MLflow** and **argparse** CLI configuration; built modular **src** structure for full reproducibility.

HTTP Server From Scratch

- **Tech Stack:** C++, CMake, Winsock
- Built a functional HTTP server in **C++** using low-level **Winsock API**, managing the full **TCP socket lifecycle**; modular architecture with **CMake** build management.

EXPERIENCE

M.Tech Research Scholar - Generative AI & Computer Vision

Sep 2024 – Present

Malaviya National Institute of Technology (MNIT), Jaipur

- Conducting thesis research on diffusion-based generative models for computer vision, working with **Stable Diffusion**, **ControlNet**, **DDPM**, **LoRA**, and **IP-Adapter** for domain-specific image synthesis and data augmentation.

Representative of Asian Students

Aug 2021 – Jul 2023

DTU International Students' Society (DTU-ISS), Delhi - Represented Asian students in the Chamber of Representatives; coordinated onboarding for newly admitted international students across two consecutive years.

EDUCATION

M.Tech, Computer Science and Engineering

2024 – Present

Malaviya National Institute of Technology (MNIT), Jaipur

Thesis: Diffusion-based generative models applied to computer vision problems

B.Tech, Computer Engineering

2020 – 2024

Delhi Technological University (DTU), Delhi

CGPA: 7.73

Thesis: Multimodal Sentiment Analysis using Late Fusion LSTM

PUBLICATIONS

Pranesh Nur, K. Khan, R. Katarya et al. *"Multimodal Sentiment Analysis using Late Fusion LSTM."* GCMM2024 International Conference, Thailand, 2024.

AWARDS & SCHOLARSHIPS

ICCR Scholarship - Indian Council for Cultural Relations, Government of India

2020, 2024

Merit scholarship for international South Asian students at premier Indian institutions. Awarded **twice**

consecutively - B.Tech at DTU (2020–2024) and M.Tech at MNIT Jaipur (2024–Present).